Planning and tool use with language models



AI agents · University · Lecture 2

In this lecture

- Plan with A*.
- Build ReAct agents.
- Ground answers in tools.

Remember

- Admissible heuristics.
- Thought, action, observation.
- Check every answer.

Search, then act

1. Warm-up (5 min): What went wrong?
2. Trace (20 min): A*.
3. Debug (15 min): ReAct.
4. Compare (15 min): Guarantees.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. When is A* optimal?
2. Why did the agent skip the tool?
3. How will you grade transcripts?

Exit ticket

- Write one instruction that grounds an agent’s answers.